HDOC DAY–6 ACTIVITY

C PROGRAMMING ASSIGNMENT REPORT

QUESTION 1 – Square Measurements

Write a C program to find the area in square meters, perimeter in meters, and diagonal in meters of a square when its side is given in millimeters.

1. Problem Understanding

- Input: Side of the square in millimeters (mm).
- Output: Area in square meters (m^2), perimeter in meters (m), and diagonal in meters (m).
- Convert millimeters to meters using: side_m = side_mm / 1000.
- Area = side_m × side_m.
- Perimeter = 4 × side_m.
- Diagonal = side_m × √2.

Formula Summary

Quantity	Formula
Side in meters	s = side_mm / 1000
Area	$A = s^2$
Perimeter	P = 4s
Diagonal	$D = s\sqrt{2}$

2. Standard Test Cases

Test Case	Input	Expected Output	Actual Output	Result
1	1000 mm	Area 1.0000 m^2; Perimeter 4.0000 m; Diagonal 1.4142 m	Area 1.0000 m^2; Perimeter 4.0000 m; Diagonal 1.4142 m	PASS
2	2500 mm	Area 6.2500 m^2; Perimeter 10.0000 m; Diagonal 3.5355 m	Area 6.2500 m^2; Perimeter 10.0000 m; Diagonal 3.5355 m	PASS
3	5000 mm	Area 25.0000 m^2; Perimeter 20.0000 m; Diagonal 7.0711 m	Area 25.0000 m^2; Perimeter 20.0000 m; Diagonal 7.0711 m	PASS

3. Algorithm

1. Start the program.
2. Declare variables for side in millimeters, side in meters, area, perimeter, and diagonal.

3. Read the side in millimeters.
4. Convert the side to meters by dividing by 1000.
5. Calculate area as side × side.
6. Calculate perimeter as 4 × side.
7. Calculate diagonal as side × $\sqrt{2}$.
8. Display the three results to four decimal places.
9. Stop the program.

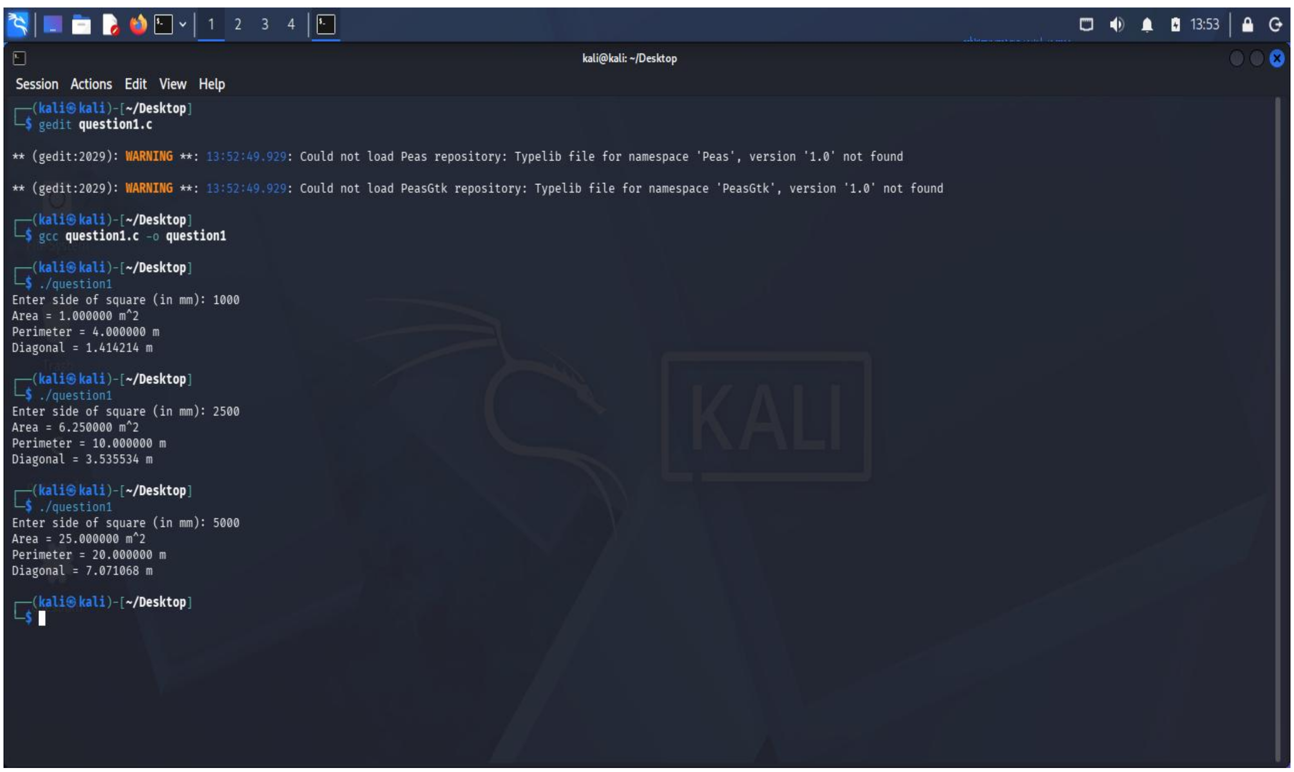
4. C Program

```
#include<stdio.h>

#include<math.h>

int main(void)

{

double side_mm, side_m, area, perimeter, diagonal;

printf("Enter side of square (in mm): ");

scanf("%lf", &side_mm);

side_m = side_mm / 1000.0;

area = side_m*side_m;

perimeter = 4.0 * side_m;

diagonal = side_m * sqrt(2.0);

printf("Area = %f m^2\n", area);

printf("Perimeter = %f m\n", perimeter);

printf("Diagonal = %f m\n", diagonal);

return 0;

}
```

5. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	1000 mm	Area = 1.000000 m^2; Perimeter = 4.000000 m; Diagonal = 1.414214 m	Area = 1.000000 m^2; Perimeter = 4.000000 m; Diagonal = 1.414214 m	PASS
2	2500 mm	Area = 6.250000 m^2; Perimeter = 10.000000 m; Diagonal = 3.535534 m	Area = 6.250000 m^2; Perimeter = 10.000000 m; Diagonal = 3.535534 m	PASS
3	5000 mm	Area = 25.000000 m^2; Perimeter = 20.000000 m; Diagonal = 7.071068 m	Area = 25.000000 m^2; Perimeter = 20.000000 m; Diagonal = 7.071068 m	PASS

6. Compilation and Execution Evidence



QUESTION 2 – Quadrant-Shaped Garden

Write a C program to find the fencing cost and area of a quadrant-shaped garden when its radius and fencing rate per meter are given.

1. Problem Understanding

- Input: Radius of the quadrant in meters and fencing rate per meter.
- Output: Total fencing length, fencing cost, and area of the quadrant.
- A quadrant is one-fourth of a circle.

- Quarter-circle arc length = $\pi r/2$.
- Total boundary to be fenced = quarter-circle arc + two radii = $\pi r/2 + 2r$.
- Fencing cost = total fencing length × fencing rate.
- Area of quadrant = $\pi r^2/4$.
- The program uses $\pi = 3.141592653589793$.

Formula Summary

Quantity	Formula
Arc length	$L_{arc} = \pi r/2$
Total fencing length	$L = \pi r/2 + 2r$
Fencing cost	Cost = L × rate
Quadrant area	$A = \pi r^2/4$

2. Standard Test Cases

Test Case	Input	Expected Output	Actual Output	Result
1	r = 7 m; Rate = 50/m	Length 24.9956 m; Cost 1249.7787; Area 38.4845 m^2	Length 24.9956 m; Cost 1249.7787; Area 38.4845 m^2	PASS
2	r = 10 m; Rate = 75/m	Length 35.7080 m; Cost 2678.0972; Area 78.5398 m^2	Length 35.7080 m; Cost 2678.0972; Area 78.5398 m^2	PASS
3	r = 14 m; Rate = 100/m	Length 49.9911 m; Cost 4999.1149; Area 153.9380 m^2	Length 49.9911 m; Cost 4999.1149; Area 153.9380 m^2	PASS

3. Algorithm

1. Start the program.
2. Define the value of π.
3. Declare variables for radius, rate, fencing length, fencing cost, and area.
4. Read the radius in meters.
5. Read the fencing rate per meter.
6. Calculate total fencing length as (π × radius / 2) + (2 × radius).
7. Calculate fencing cost as fencing length × rate.
8. Calculate quadrant area as π × radius^2 / 4.
9. Display fencing length, fencing cost, and area to four decimal places.
10. Stop the program.

4. C Program

```
#include <stdio.h>

#define PI 3.141592653589793
```

```c
int main(void)

{

double radius, rate, fencing_length, fencing_cost, area;

printf("Enter radius of quadrant (in m): ");

scanf("%lf", &radius);

printf("Enter fencing rate (per m): ");

scanf("%lf", &rate);

fencing_length = (PI * radius / 2.0) + (2.0 * radius);

fencing_cost = fencing_length * rate;

area = (PI * radius * radius) / 4.0;

printf("Fencing length = %f m\n", fencing_length);

printf("Fencing cost = %f\n", fencing_cost);

printf("Area = %f m^2\n", area);

return 0;

}
```

5. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	r = 7 m; Rate = 50/m	Length = 24.995574 m; Cost = 1249.778702; Area = 38.484510 m^2	Length = 24.995574 m; Cost = 1249.778702; Area = 38.484510 m^2	PASS
2	r = 10 m; Rate = 75/m	Length = 35.707963 m; Cost = 2678.097245; Area = 78.539816 m^2	Length = 35.707963 m; Cost = 2678.097245; Area = 78.539816 m^2	PASS
3	r = 14 m; Rate = 100/m	Length = 49.991148 m; Cost = 4999.114802; Area = 153.938040 m^2	Length = 49.991148 m; Cost = 4999.114802; Area = 153.938040 m^2	PASS

6. Compilation and Execution Evidence